Fiscal Year:	FY 2017	**Task Last Updated:**	FY 06/28/2017
PI Name:	Kiss, John Z Ph.D.		
Project Title:	Novel Explorations Into the Interactions Between Light and Gravity Sensing in Plants		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	27402-6170	**Congressional District:**	13
Comments:	NOTE: PI moved to Florida Institute of Technology from University of North Carolina-Greensboro in May 2024 (Ed., 2/6/25). NOTE: PI moved to University of North Carolina-Greensboro from University of Mississippi in July 2016 (Ed., 6/13/17) NOTE: PI moved to U Miss from Miami (Ohio) University in September 2012 (Ed., 8/10/2012)		
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Space Life Sciences-Flight NNH09ZTT002N
Start Date:	09/11/2012	**End Date:**	08/09/2017
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 8/9/2017 when new continuation grant issued; information per F. Hernandez/ARC (Ed., 12/11/17) NOTE: End date changed to 9/30/2017 per F. Hernandez/ARC (Ed., 1/18/17) NOTE: End date is now 12/31/2016, per NSSC information (Ed., 6/15/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Edelmann, Richard Ph.D. (Miami University)		
Grant/Contract No.:	NNX12AO65G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	[ED. NOTE 9/28/2012: Continuation of "Novel Explorations Into the Interactions Between Light and Gravity Sensing in Plants"--grant # NNX10AM86G, per PI move from Miami-Ohio to University of Mississippi. See that project for previous reports] The major goals of this project are to determine how gravity and light responses influence each other in plants and to better understand the cellular signaling mechanisms involved in plant tropisms. This proposed project builds on our previous spaceflight experiments on the International Space Station (ISS) with the European Modular Cultivation System (EMCS) using the model plant Arabidopsis. In this new proposed project, we plan to confirm and extend our discovery during the recent ISS experiment of a novel red-light-based positive phototropic response in plant shoots. Our hypothesis is that positive red-light-sensing, which was known in older plant lineages, is masked by normal 1-g conditions in more recently evolved lineages. We also propose to investigate fundamental interactions among red and blue light signaling pathways and the gravity sensing mechanisms. The ISS is the only laboratory facility where it is possible to reliably and simultaneously vary light and gravity stimuli for the proposed experiments. These experiments can be accomplished with telemetric science and therefore fit with the extremely limited down mass capabilities of current spaceflight opportunities. The EMCS is an automated facility and can be operated largely with telemetry. Experimental containers will launched via Space-X, and during an experimental run, images of seedling growth responses can be downlinked to Earth. A significant advantage of this proposed project is that we will use existing flight hardware that has been flown successfully and allowed for growth of Arabidopsis seedlings in the EMCS. This project is also relevant to the recent emphasis at NASA to study plant growth and development at fractional g-levels such as those found on the moon and Mars. Improved knowledge of the basic mechanistic processes that will be the focus of this project is vital to develop ways to use plants in extraterrestrial bioregenerative life support systems. EDITOR'S NOTE (December 2017): Project is being continued by "Novel Explorations Into The Interactions Between Light and Gravity Sensing in Plants--80NSSC17K0546" with the same PI, Dr. John Kiss, due to Principal Investigator's move to University of North Carolina at Greensboro.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We plan to study the effects of gravity and light on basic cellular mechanisms in plants, so this project is relevant to the emphasis on using Arabidopsis thaliana as a model system in developmental biology. Since plants will be a necessary part of bioregenerative life support needed to send humans to Mars and beyond, the knowledge obtained from our spaceflight experiments will be critical for developing ways to effectively use plants in the bioregenerative life support systems. These life support systems also can be used in ground-based Earth applications such as the design of large-scale greenhouses to support crop plants and agriculture. In addition, improved knowledge of light and gravity systems in plants will, in the long term, allow us to develop better crop plants for use on Earth.
Task Progress:	Summary of Major Results ***Project 1: Phototropic curvature*** Utilizing the European Modular Cultivation System (EMCS) on board the International Space Station (ISS), we investigated the interaction between phototropic and gravitropic responses in three Arabidopsis thaliana genotypes, Landsberg wild type, as well as mutants of phytochrome A and phytochrome B. Onboard centrifuges were used to create a fractional gravity gradient ranging from reduced gravity up to 1 g. A novel positive blue-light phototropic response of roots was observed during conditions of microgravity, and this response was attenuated at 0.1 g. In addition, a red-light pretreatment of plants enhanced the magnitude of positive phototropic curvature of roots in response to blue illumination. Additionally, a positive phototropic response of roots was observed when exposed to red light, and a decrease in response was gradual and correlated with the increase in gravity. The positive red-light phototropic curvature of hypocotyls when exposed to red light was also confirmed. Both red-light and blue-light phototropic responses were also shown to be a product of directional light intensity. To our knowledge, this is the first characterization of a positive blue-light phototropic response in Arabidopsis roots, as well as the description of the relationship between these phototropic responses in fractional or reduced gravity. ***Project 2: Gene Profiling*** The transcriptome of seedlings were analyzed from experiments performed on the International Space Station (ISS) to study the interacting effects of light and gravity on plant tropisms. Seeds of Arabidopsis thaliana were germinated and grown in cassettes in the European Modular Cultivation System (EMCS) for 4 days at 1g then exposed to a range of gravitational accelerations (microgravity, 0.07g, 0.13g, 0.21g, 0.39g, 0.53g, and 1g) and light treatments (blue light with or without a 1 hour pretreatment with red light). At the end of the space experiments, the cassettes containing the seedlings were frozen in the Minus Eighty Laboratory Freezer (MELFI) and returned to Earth on space shuttle STS-131. The RNA was extracted from whole seedlings and used for the transcriptome analyses. A comparison of 1g-treated-spaceflight samples with ground controls identified 230 genes that were differentially regulated at least two-fold ($p < 0.01$). A further comparison of all spaceflight samples with ground controls identified approximately 280 genes that were differentially regulated at least two fold ($p < 0.01$). Of these genes, several were involved in regulating cell polarity (i.e., auxin, calcium, lipid metabolism), cell wall development, oxygen status, and cell defense or stress. However, when the transcriptome of the all g-treated spaceflight samples were compared with microgravity samples, only approx.130 genes were identified as being differently regulated ($p < 0.01$). Of these, only 27 were at least 2 fold differently regulated between microgravity and 1g samples and included putatitive/pseudo/undefined genes (n=14), transposable elements (n=5), an expansin (ATEXP24; At1g21240), a cell-wall kinase (WAK3; At1g21240), a laccase-like flavonoid oxidase (TT10;At5g48100), among others. Not surprisingly, there was a difference in the % of genes represented in individual gene categories when comparing spaceflight samples with ground controls and spaceflight microgravity versus spaceflight g-treated space samples. These results emphasize the need for tissue fixation on the centrifuge for an accurate spaceflight control. These results are one of the few reports of transcriptome analyses from a space experiment with plants that had gravity treatments using an on-board centrifuge. EDITOR'S NOTE (December 2017): Project is being continued by "Novel Explorations Into The Interactions Between Light and Gravity Sensing in Plants--80NSSC17K0546" with the same PI, Dr. John Kiss, due to Principal Investigator's move to University of North Carolina at Greensboro. See that project for subsequent reporting.

Bibliography Type:	Description: (Last Updated: 11/25/2025)
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